

Spectral Sector Decomposition in the Rubik's Cube Representation

Block Spectral Structure and a Conditional Rationality Criterion

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Abstract

Problem. We study the averaging operator $A = \frac{1}{|S|} \sum_{s \in S} \rho(s)$ of a finite group representation, separating its block-spectral structure from possible arithmetic explanations. In the 228-dimensional Rubik's cube representation with the standard 18 face-turn generators, six eigenvalues are numerically registered against the displayed rational values.

Approach. We decouple the spectral calculation from the arithmetic question. First, block restriction reduces the canonical spectrum to four finite block computations. Second, the eigenspace compression-trace identity gives an elementary rationality criterion and a partition-integrality certificate format. The latter is not used to prove canonical Rubik rationality: the registered face partition does not supply its integer hypothesis.

Results. For the standard 18-generator realization, the computed block spectra have union $\lambda \in \{1, 8/9, 7/9, 2/3, 5/9, 1/3\}$, with multiplicities summing to 228. The cp and ep reductions are analytic. On the co block, covariance under the orientation-preserving cubic rotation group O supplies a Schur reduction, while the accidental degeneracy remains computational; the eo assignment remains a numerical-representation observation. No complex-linear O_h action is used.

Boundary. Spectral rationality is not uniform across generator choices: a registered broken-face control contains a conjugate pair numerically recognized against $\mathbb{Q}(\sqrt{5})$. This finite contrast does not establish a general arithmetic classification. The six-layer census and the conditional trace criteria remain logically independent outputs.

Notation

Symbol	Meaning
G	Finite group
$S \subset G$	Symmetric generating set ($S = S^{-1}$)
$\rho : G \rightarrow \text{GL}(V)$	Finite-dimensional unitary representation
V	Representation space (228-dim for the Rubik cube)
block	A cubie-type invariant subspace: cp (corner perm, 64-dim), ep (edge perm, 144-dim), co (corner ori, 8-dim), eo (edge ori, 12-dim)
$A = \frac{1}{ S } \sum_s \rho(s)$	Averaging operator — Hermitian when $S = S^{-1}$ and ρ is unitary
$\lambda = 1 - k/m$	Eigenvalue parametrization used for the registered families ($m = 9$ for 18 face-turn generators)
k -set	Set of k values producing distinct eigenvalues: $\{0, 1, 2, 3, 4, 6\}$ for 18-full
P_λ	Spectral projector onto eigenspace E_λ
layer $V_\lambda = \text{im}(P_\lambda)$	An eigenspace of the averaging operator A — 6 canonical layers
O, U_R	Orientation-preserving cubic rotation group and its registered complex-linear unitary action
$V_1, V_{8/9}, V_{7/9}, V_{2/3}, V_{5/9}, V_{1/3}$	Canonical layers ($\lambda = 1 - k/9, k \in \{0, 1, 2, 3, 4, 6\}$)
$\chi_\lambda(s) = \text{Tr}(P_\lambda \rho(s) P_\lambda)$	Eigenspace compression trace used in the conditional rationality criterion
Bose-Mesner algebra	Commuting algebra of a permutation association scheme; used in the cp reduction
J	Edge-face incidence matrix, with $J_{e,f} = 1$ when edge position e lies on face f

Definition 0.1 (Spectral Layer).

An eigenspace $V_\lambda = \text{im}(P_\lambda)$ of the averaging operator A . The 228-dimensional Rubik’s cube representation has 6 canonical layers: $V_1, V_{8/9}, V_{7/9}, V_{2/3}, V_{5/9}, V_{1/3}$, corresponding to $\lambda = 1 - k/9$ with $k \in \{0, 1, 2, 3, 4, 6\}$.

1 Introduction

Averaging group elements in a finite-dimensional representation often produces substantial spectral collapse. In the Rubik’s cube setting, the averaging operator built from the standard 18 face-turn generators has six computed eigenvalues, parametrized by $\lambda = 1 - k/9$ with $k \in \{0, 1, 2, 3, 4, 6\}$. The first question is structural: how does the invariant cubie-type decomposition reduce this spectrum? The second is arithmetic: what additional hypothesis would force rational eigenvalues? These questions are independent. Block diagonality determines how restricted spectra combine, but it does not determine their arithmetic field.

The arithmetic statement is elementary and conditional. Specifically, the compression-trace identity renders eigenvalue rationality equivalent to the rationality of a single total trace sum. **Partition integrality** is only a stronger certificate format: if a predeclared partition has integer compression-trace sums on every part, then the total sum is rational. In the registered canonical computation, however, the Rubik face partition does not supply this integer hypothesis. Ordinary generator traces and eigenspace compression traces must therefore remain distinct.

Four claim-status levels. The paper distinguishes four claim levels:

1. **Theorem.** Block spectral union and the conditional trace criteria are exact under their stated hypotheses.
2. **Computational Certificate.** The registered block k -sets, analytic reductions, and computational inputs are reported and labelled separately below.
3. **Computational Observation.** The broken-face control provides a finite numerical contrast rather than a general arithmetic classification.

4. **Research Program.** A unified exact arithmetic explanation of all four canonical blocks remains open.

Finite-family contrast. A registered broken-face control contains conjugate values numerically recognized in $\mathbb{Q}(\sqrt{5})$. Thus the ambient representation data alone do not determine the observed spectral field. This is a computational negative control, not a general symmetry theorem.

The paper first proves the general averaging and block-union theorems, then records the canonical Rubik census with its per-block proof status. It next establishes the conditional trace criteria, reports one finite negative control, and closes with an explicit theorem-computation boundary. The blockwise output can serve as spectral input to finer joint-sector analyses, while the arithmetic criterion remains a separate conditional layer.

2 Setting and notation

2.1 Concrete conventions

We use a right-handed Cartesian coordinate system: $+X \rightarrow R$, $+Y \rightarrow U$, $+Z \rightarrow F$. Corner positions are indexed by sign vectors $\{x \in \{\pm 1\}^3 : \prod_i x_i \neq 0\}$; edge positions by vectors in $\{x \in \{\pm 1, 0\}^3 : \sum_i |x_i| = 2\}$. The accompanying computational artifacts record the face mapping, generator encoding, and array ordering used throughout the computational case study.

The representation space $V = \mathbb{C}^{228}$ decomposes into four G -invariant blocks:

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}}, \quad 64 + 144 + 8 + 12 = 228,$$

laid out in $\text{cp} \rightarrow \text{ep} \rightarrow \text{co} \rightarrow \text{eo}$ order. The generator set S is the 18 standard face-turn generators ($S = S^{-1}$).

2.2 Abstract setting

Let G be a finite group and let $\rho : G \rightarrow \text{GL}(V)$ be a finite-dimensional unitary representation. Let $S \subseteq G$ be a finite inverse-closed generating set, meaning $S = S^{-1}$. Define the averaging operator

$$A = \frac{1}{|S|} \sum_{s \in S} \rho(s) \in \text{End}(V).$$

Proposition 2.1 (Inverse-Closure Hermiticity).

For a unitary representation ρ , inverse closure $S = S^{-1}$ is sufficient for the averaging operator $A = \frac{1}{|S|} \sum_{s \in S} \rho(s)$ to be Hermitian.

Proof. Since ρ is unitary, $\rho(s)^\dagger = \rho(s^{-1})$ for every $s \in G$. The adjoint of A is

$$A^\dagger = \frac{1}{|S|} \sum_{s \in S} \rho(s)^\dagger = \frac{1}{|S|} \sum_{s \in S} \rho(s^{-1}) = \frac{1}{|S|} \sum_{t \in S^{-1}} \rho(t).$$

Hence $S = S^{-1}$ implies $A^\dagger = A$. The converse need not hold for an arbitrary representation because distinct group-algebra coefficients can have the same represented operator. $\square$

Boundary note. *For the Rubik's cube, the standard 18 face-turn generators satisfy $S = S^{-1}$ with equal weights, so A is Hermitian and admits an orthogonal spectral decomposition. Non-inverse-closed families require a separate normality or nonnormal spectral analysis.*

We write E_λ for the eigenspace of A with eigenvalue λ , and P_λ for the orthogonal projector onto E_λ .

3 Main results

Proposition 3.1 (Eigenspace Compression-Trace Identity).

For every eigenvalue λ of A with orthogonal projector P_λ onto E_λ ,

$$\lambda = \frac{1}{d_\lambda} \cdot \frac{1}{|S|} \sum_{s \in S} \chi_\lambda(s), \quad \chi_\lambda(s) := \text{Tr}(P_\lambda \rho(s) P_\lambda), \quad d_\lambda = \dim E_\lambda.$$

Terminology — eigenspace compression trace. The quantity $\chi_\lambda(s)$ is the trace on E_λ of the compression $P_\lambda \rho(s) P_\lambda$. It is not the character of a restricted subrepresentation unless E_λ is $\rho(G)$ -invariant. Cyclicity and $P_\lambda^2 = P_\lambda$ give $\text{Tr}(P_\lambda \rho(s) P_\lambda) = \text{Tr}(P_\lambda \rho(s))$.

Proof. Since $AP_\lambda = \lambda P_\lambda$, taking traces gives

$$\text{Tr}(AP_\lambda) = \lambda \cdot \text{Tr}(P_\lambda) = \lambda d_\lambda.$$

On the other hand, by linearity of the trace,

$$\text{Tr}(AP_\lambda) = \frac{1}{|S|} \sum_{s \in S} \text{Tr}(P_\lambda \rho(s) P_\lambda) = \frac{1}{|S|} \sum_{s \in S} \chi_\lambda(s).$$

Combining the two identities yields the claim. $\square$

Theorem 3.2 (Block Compatibility Lemma).

Let A be Hermitian on a finite-dimensional Hilbert space, and let

$$V = \bigoplus_{i=1}^k V_i,$$

be an orthogonal decomposition with block projectors P_i . If $[P_i, A] = 0$ for every i , then each spectral projector P_λ of A satisfies

$$P_\lambda = \bigoplus_{i=1}^k P_{\lambda,i}, \quad P_{\lambda,i} := P_i P_\lambda P_i.$$

Equivalently, P_λ commutes with every block projector: $[P_\lambda, P_i] = 0$ for all i .

Proof. The commutation hypotheses imply that A is block diagonal: $A = \bigoplus_i A_i$, where $A_i = A|_{V_i}$. By Lagrange interpolation on the distinct eigenvalues, P_λ is a polynomial in A . It therefore commutes with every P_i and is block diagonal, giving the stated formula. $\square$

Rubik specialization. Each cubie-type block is $\rho(G)$ -invariant, so its projector commutes with every $\rho(g)$ and hence with the average A .

Corollary 3.3.

For each block V_i , the restricted projector $P_{\lambda,i}$ is an orthogonal projector on V_i satisfying $P_{\lambda,i}^2 = P_{\lambda,i}$, $P_{\lambda,i}^* = P_{\lambda,i}$, and $P_{\lambda,i} A_i = A_i P_{\lambda,i} = \lambda P_{\lambda,i}$ whenever $P_{\lambda,i} \neq 0$.

Role of the lemma.

Theorem 3.2 is a direct consequence of block diagonality. Its value is organizational: it reduces the 228-dimensional problem to four independent sub-problems and shows that every global spectral projector restricts blockwise.

3.1 Block Spectral Union and Canonical Census

Theorem 3.4 (Block Spectral Union).

If $V = \bigoplus_B V_B$ and every V_B is A -invariant, then

$$\text{Spec}(A) = \bigcup_B \text{Spec}(A_B), \quad A_B = A|_{V_B}.$$

This is a set-theoretic union. If the same eigenvalue occurs in several blocks, the corresponding global eigenspace is the direct sum of those block eigenspaces.

Proof. The orthogonal block decomposition implies $A = \bigoplus_B A_B$. Therefore, the characteristic polynomial factors as $\det(tI - A) = \prod_B \det(tI_B - A_B)$, yielding the claimed spectral union. $\square$

Computational Proposition 3.5 (Canonical Block Census). *For the registered 18-generator Rubik realization, the four block spectra are:*

Block	dim	k -set in $\lambda = 1 - k/9$	multiplicities by k	Status
cp	64	$\{0, 4, 6\}$	$(8, 24, 32)$	analytic Hamming-scheme reduction
ep	144	$\{0, 2, 3, 4\}$	$(12, 36, 24, 72)$	analytic face-incidence reduction
co	8	$\{3, 4, 6\}$	$(2, 3, 3)$	symmetry-guided computation; one accidental degeneracy remains numerical
eo	12	$\{1, 2, 4\}$	$(2, 3, 7)$	numerical-representation observation

Their union is $\{0, 1, 2, 3, 4, 6\}$, giving six global layers. The block contributions are:

Global layer	λ	k	dim	cp	ep	co	eo
V_1	1	0	20	8	12	—	—
$V_{8/9}$	8/9	1	2	—	—	—	2
$V_{7/9}$	7/9	2	39	—	36	—	3
$V_{2/3}$	2/3	3	26	—	24	2	—
$V_{5/9}$	5/9	4	106	24	72	3	7
$V_{1/3}$	1/3	6	35	32	—	3	—

The dimensions sum to 228. The value $k = 5$ is absent from every registered block spectrum. This proposition is a finite canonical census; it does not identify a universal algebraic factorization.

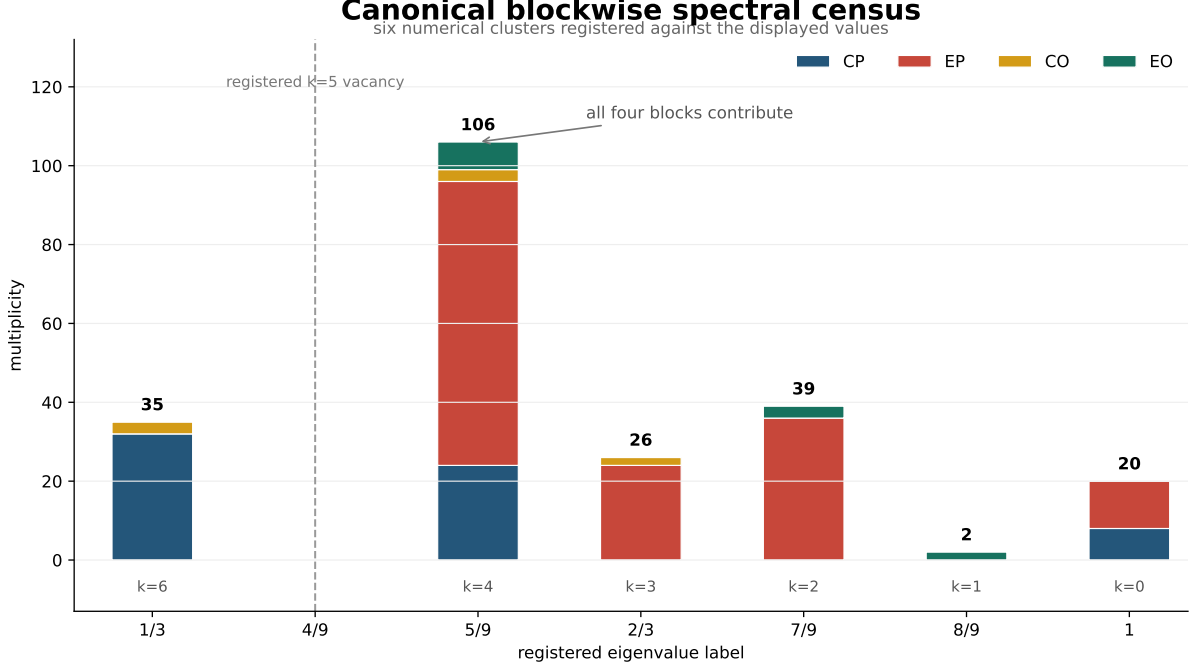


Figure 1. The six canonical spectral layers of the averaging operator, showing eigenvalue $\lambda = 1 - k/9$, dimension, and block support. The figure records the canonical block census; it does not assert a general rationality mechanism.

4 Per-block spectral analysis

The 228-dimensional space decomposes into four $\rho(G)$ -invariant blocks:

$$V = V_{\text{cp}} \oplus V_{\text{ep}} \oplus V_{\text{co}} \oplus V_{\text{eo}},$$

with dimensions 64, 144, 8, and 12, respectively. By Theorem 3.2, each eigenspace projector splits as

$$P_\lambda = P_{\lambda,\text{cp}} \oplus P_{\lambda,\text{ep}} \oplus P_{\lambda,\text{co}} \oplus P_{\lambda,\text{eo}}.$$

The four restricted spectra do not have the same proof status. The following summary is therefore part of the claim boundary, not merely an implementation detail.

4.1 Permutation blocks

The generators act by permutation matrices on V_{cp} and V_{ep} . Both spectra admit exact combinatorial reductions on the position factors.

4.1.1 Corner-permutation reduction

Write

$$\rho_{\text{cp}}(s) = P_8(s) \otimes I_8, \quad S_8 = \sum_{s \in S} P_8(s).$$

Identify the eight corner positions with the binary hypercube Q_3 , and let A_r be its distance- r adjacency matrix. Entrywise counting of the 18 face turns yields

$$S_8 = 9I + 2A_1 + A_2.$$

Indeed, a corner has nine fixed contributions; distance-one corners receive two transitions, distance-two corners one, and antipodal corners none. For $u \in \{0, 1\}^3$, the character $v_u(x) = (-1)^{u \cdot x}$ is a common eigenvector. The eigenvalues depend only on $w = |u|$:

w	multiplicity	A_1	A_2	S_8
0	1	3	3	18
1	3	1	-1	10
2	3	-1	-1	6
3	1	-3	3	6

Therefore

$$\text{Spec}(A_{\text{cp}}) = \{1^{(8)}, (5/9)^{(24)}, (1/3)^{(32)}\}, \quad A_{\text{cp}} = \frac{1}{18} S_8 \otimes I_8.$$

4.1.2 Edge-permutation reduction

Let $J \in \{0, 1\}^{12 \times 6}$ be the edge-face incidence matrix. Every edge belongs to two faces and every face contains four edges. With $\rho_{\text{ep}}(s) = P_{12}(s) \otimes I_{12}$, entrywise counting gives

$$S_{12} := \sum_{s \in S} P_{12}(s) = 10I + JJ^*, \quad J^*J = 4I + A_{\text{oct}},$$

where A_{oct} is the adjacency matrix of the octahedron graph on the six faces. Since

$$\text{Spec}(A_{\text{oct}}) = \{4^{(1)}, 0^{(3)}, (-2)^{(2)}\},$$

the nonzero spectrum of JJ^* is $\{8^{(1)}, 4^{(3)}, 2^{(2)}\}$, with an additional six-dimensional kernel. Hence

$$\text{Spec}(S_{12}) = \{18^{(1)}, 14^{(3)}, 12^{(2)}, 10^{(6)}\},$$

and therefore

$$\text{Spec}(A_{\text{ep}}) = \{1^{(12)}, (7/9)^{(36)}, (2/3)^{(24)}, (5/9)^{(72)}\}, \quad A_{\text{ep}} = \frac{1}{18} S_{12} \otimes I_{12}.$$

These exact finite reductions justify the analytic/combinatorial status of the permutation blocks. The orientation blocks below retain different claim statuses.

4.2 Corner-orientation block (co)

The corner-orientation block carries non-real phase entries. The generators act by matrices over $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$:

$$\rho_{\text{co}}(s) \in M_8(\mathbb{Z}[\omega]) \subset M_8(\mathbb{Q}(\omega)).$$

Computational Proposition 4.1 (Corner-Orientation Spectrum). For the registered 18-generator realization,

$$\text{Spec}(A_{\text{co}}) = \{(2/3)^{(2)}, (5/9)^{(3)}, (1/3)^{(3)}\}.$$

The permutation action of the orientation-preserving cubic rotation group O on the eight corners decomposes as $A_1 \oplus A_2 \oplus T_1 \oplus T_2$. The registered rephased unitaries U_R satisfy $[U_R, A_{\text{co}}] = 0$ to numerical precision, so this multiplicity-free decomposition supplies a Schur reduction. No complex-linear action of the full reflection group O_h is used.

For $M_{\text{co}} = 18(A_{\text{co}} - I/2)$, the registered calculation gives

$$\text{Spec}(M_{\text{co}}) = \{3^{(2)}, 1^{(3)}, -3^{(3)}\}$$

and hence the displayed spectrum. The equality of the two one-dimensional eigenvalues is not forced by the O -module dimensions and remains a numerical input. Similarly, the two three-dimensional irrep labels are not explicitly assigned. Thus the Schur reduction is structural, but the complete eigenvalue assignment is a computational proposition.

Complete-face phase cancellation

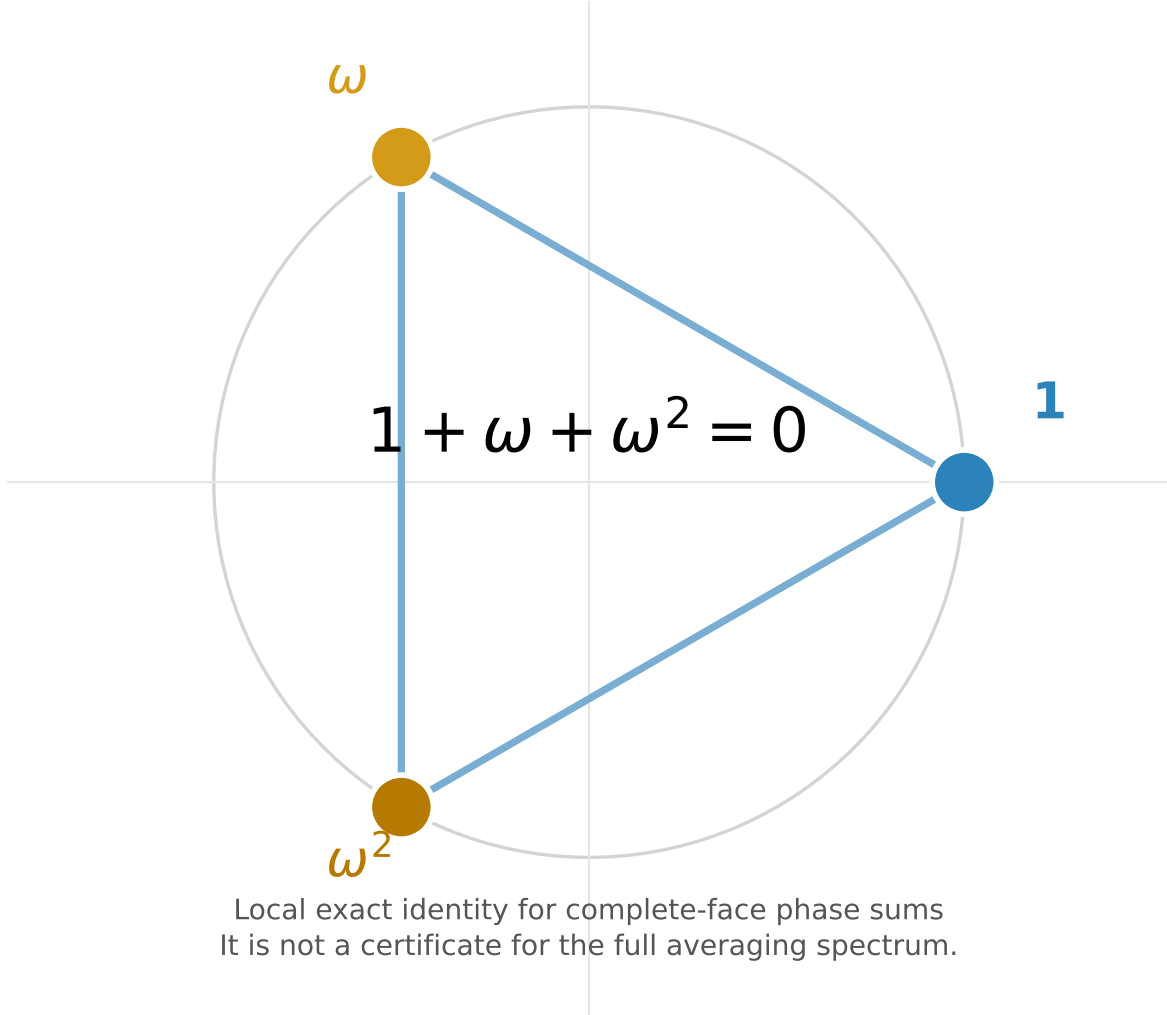


Figure 2. The three cube roots of unity and the identity $\omega + \omega^2 + 1 = 0$. The figure illustrates phase cancellation on a complete face; it does not assert compression-trace integrality or prove rationality of the full averaging spectrum.

4.3 Edge-orientation block

Computational Observation 4.2 (Edge-Orientation Spectrum). The registered 12×12 signed-permutation average has

$$\text{Spec}(A_{\text{eo}}) = \{(8/9)^{(2)}, (7/9)^{(3)}, (5/9)^{(7)}\}.$$

The trace check is $2(8/9)+3(7/9)+7(5/9) = 8 = \text{Tr}(A_{\text{eo}})$. The edge permutation representation contains a multiplicity-two T_2 component, so rotation covariance alone permits a non-scalar multiplicity matrix and does not determine the three eigenvalues. This block therefore remains a numerical-representation observation.

4.4 Computational Projector Certificate

Computational Certificate 4.3 (Canonical Projector Residuals). In the registered complex128 realization, the six numerical spectral projectors have maximum idempotence residual 1.09×10^{-14} , maximum pairwise orthogonality residual 4.09×10^{-15} , and completeness residual 1.70×10^{-14} in Frobenius norm. Individual compression traces $\text{Tr}(P_\lambda \rho(s) P_\lambda)$ can be complex.

This observation is basis-dependent and makes no algebraic classification claim. In particular, rational eigenvalues do not justify treating the computed projectors or individual compression traces as entrywise rational in the declared realization.

5 Generator traces

This section concerns ordinary traces of the representation blocks. These are not the compression traces $\chi_\lambda(s) = \text{Tr}(P_\lambda \rho(s) P_\lambda)$ used below.

$$\rho(g) = P_{\text{cp}}(g) \oplus P_{\text{ep}}(g) \oplus \Omega_{\text{co}}(g) \oplus \Sigma_{\text{eo}}(g).$$

Computational Proposition 5.1 (Registered Generator Traces). For each of the 18 registered face-turn generators s , the ordinary block traces

$$\tau_B(s) := \text{Tr}(\rho_B(s)), \quad B \in \{\text{cp}, \text{ep}, \text{co}, \text{eo}\},$$

are integers.

Proof. The permutation-block traces count fixed basis states. The signed edge-orientation block has integer trace. In the corner-orientation block, the ω and ω^2 contributions occur with balanced multiplicities for each canonical face turn, leaving an integer trace. $\square$

Computational Corollary 5.2 (Ordinary Face-Triple Trace). For a canonical face triple $\{g, g^{-1}, g_{180}\}$,

$$\text{Tr}(\rho(g) + \rho(g^{-1}) + \rho(g_{180})) \in \mathbb{Z}.$$

This corollary concerns the ordinary trace on V . It does not imply that $\sum_s \text{Tr}(P_\lambda \rho(s))$ is integral on each eigenspace.

6 Conditional arithmetic criteria

Corollary 6.1 (Trace Rationality Criterion).

Let S be finite, let

$$A = \frac{1}{|S|} \sum_{s \in S} \rho(s).$$

For every eigenvalue λ of A ,

$$\lambda \in \mathbb{Q} \iff \sum_{s \in S} \chi_\lambda(s) \in \mathbb{Q}.$$

Proof. Proposition 3.1 gives

$$\sum_{s \in S} \chi_\lambda(s) = |S|d_\lambda \lambda.$$

Since $|S|d_\lambda$ is a positive integer, either side is rational exactly when the other is rational. $\square$

Proposition 6.2 (Partition-Integrality Certificate).

Fix an eigenvalue λ and a predeclared partition $S = \bigsqcup_{i=1}^k S_i$. If

$$\sum_{s \in S_i} \chi_\lambda(s) \in \mathbb{Z}, \quad i = 1, \dots, k,$$

then $\lambda \in \mathbb{Q}$.

Proof. Summing the integer part-sums produces an integer total sum, which is therefore rational, so Corollary 6.1 applies. $\square$

Boundary. *This proposition is a sufficient certificate format. It does not assert that such a partition exists, and the partition has no further role in the proof beyond decomposing the total sum.*

Computational Observation 6.3 (Canonical Face-Partition Audit). *In the declared canonical realization, individual compression traces can be complex, and the computed per-face sums are not all integers. In particular, the symmetric face sums for $V_{8/9}$ and $V_{5/9}$ are $16/3$ and $530/3$, respectively. Thus, the ordinary generator-trace integrality established in Computational Proposition 5.1 does not supply the hypothesis required by Proposition 6.2.*

7 Finite Generator-Family Contrast

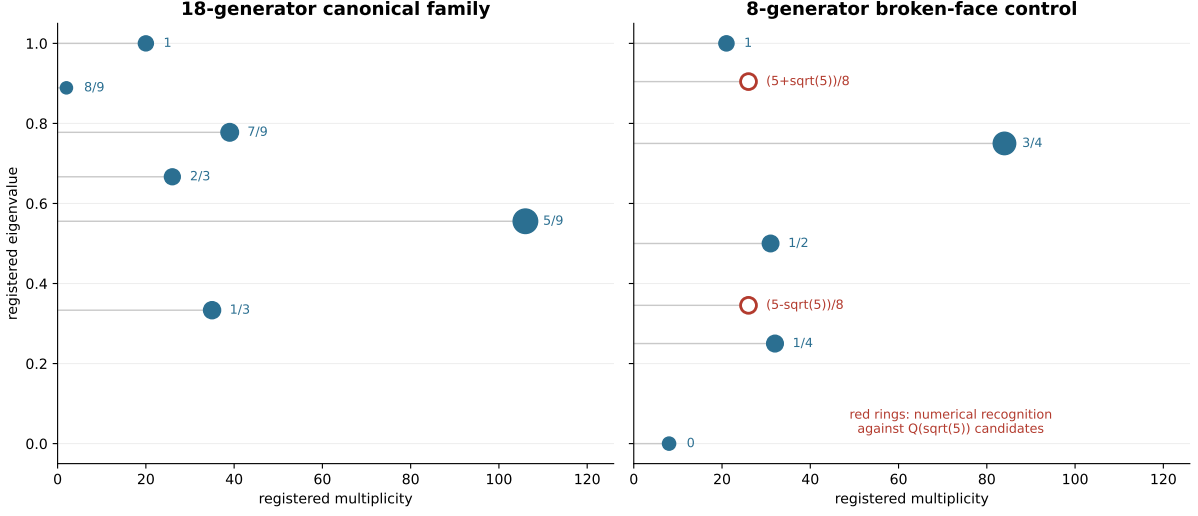
The block-union theorem alone does not determine the arithmetic field of a modified averaging operator. A finite negative control makes that boundary explicit.

Computational Observation 7.1 (Registered Broken-Face Control). In the registered generator family containing eight quarter turns, formed by removing one spatial axis, the computed spectrum contains

$$\lambda_\pm = \frac{5 \pm \sqrt{5}}{8} \approx 0.90450850, 0.34549150.$$

These values occur in the ep and eo restrictions. Their displayed quadratic forms are numerical recognitions against the declared matrices, not exact minimal-polynomial certificates. The observation shows only that changing the generator family can change the observed spectral field. It does not make face completeness necessary or sufficient for rationality. A broader finite generator-family census is available as supplementary data.

Registered generator-family arithmetic contrast



The contrast records two finite computations; it is not an exact field-classification theorem.

Figure 3. Computed contrast between the registered 18-generator family and the registered eight-generator control. Two eigenvalues are registered numerically against $\mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}$; no exact minimal-polynomial claim is made.

8 Discussion

The paper has two independent outputs. Block compatibility reduces the canonical averaging operator to four restrictions, and the registered census matches six eigenvalues to the displayed rational values. Separately, the compression-trace identity gives an elementary equivalence between eigenvalue rationality and rationality of the total compression-trace sum. Partition integrality is only a stronger certificate format. Because the tested canonical face partition fails this integer hypothesis, partition integrality does not explain the six registered spectral values.

The blockwise census also interfaces with a finer joint-spectral construction. Let QT_{all} be the uniform average of the twelve quarter turns and HT_{all} the uniform average of the six half turns. Then, exactly by definition,

$$A_{18} = \frac{2}{3}QT_{\text{all}} + \frac{1}{3}HT_{\text{all}}.$$

If these two Hermitian averages commute exactly, the images of the primitive spectral idempotents of their generated commutative algebra form a joint spectral resolution. The QT/HT joint-sector registration and its conditional collision-quotient interpretation are studied separately [3, 2]; neither is used in the proofs here.

9 Claim Status and Boundary

The general statements concern finite-dimensional unitary representations, finite inverse-closed generator sets, orthogonal reducing decompositions, and the conditional trace criteria. The six layers, their multiplicities, the missing $k = 5$, and the broken-face control are realization-specific.

We distinguish four claim levels: Theorem, Computational Certificate, Computational Observation, and Research Program. Mathematical labels such as proposition and corollary remain result types inside the Theorem level.

9.1 Theorem

- **Inverse-closure Hermiticity:** inverse closure is a sufficient condition for a unitary average to be Hermitian.
- **Compression-trace identity:** every eigenvalue satisfies the stated projector-trace identity.
- **Block compatibility and spectral union:** an orthogonal decomposition whose block projectors commute with A reduces the spectrum to the set-theoretic union of block spectra.
- **Conditional trace criteria:** eigenvalue rationality is equivalent to rationality of the total compression-trace sum; per-part integrality is a stronger sufficient certificate.

9.2 Computational Certificate

- The registered 18-generator realization has six numerical layers matched to the displayed rational values and the block multiplicities in Proposition 3.5.
- The cp and ep spectra have analytic combinatorial reductions; the co accidental degeneracy and eo spectrum retain their stated computational qualifications.
- The registered generator matrices have integer block traces. These are ordinary traces, not compression traces.
- The canonical numerical projectors satisfy the displayed residual bounds.

9.3 Computational Observation

- The eight-generator control has two values numerically recognized in $\mathbb{Q}(\sqrt{5})$.

9.4 Research Program

The canonical Rubik rationality still lacks a single exact arithmetic theorem covering all four blocks under the present hypotheses. Exact co/eo certificates, generator families satisfying partition integrality, and any nontrivial structural rationality mechanism remain open.

10 Conclusion

Inverse closure makes the averaging operator Hermitian, invariant physical blocks reduce its spectrum to a blockwise union, and the registered Rubik realization yields six layers with the displayed multiplicities. The permutation blocks have analytic reductions; the orientation blocks retain their stated computational qualifications. Partition integrality is a valid sufficient certificate format, but it is not satisfied by the tested canonical face partition and is not used to promote the numerical census.

Appendix A Canonical Spectral Summary

Compact reference for the six canonical layers. The exact permutation-block derivations appear in Section 4. The finite verification scripts and structured repository records provide the reproducibility material for the registered census.

A.1 Six Canonical Layers

k	$\lambda = 1 - k/9$	Dim	Label	Block composition
0	1	20	V_1	$\text{cp}(8) + \text{ep}(12)$
1	$8/9$	2	$V_{8/9}$	$\text{eo}(2)$
2	$7/9$	39	$V_{7/9}$	$\text{ep}(36) + \text{eo}(3)$
3	$2/3$	26	$V_{2/3}$	$\text{ep}(24) + \text{co}(2)$
4	$5/9$	106	$V_{5/9}$	$\text{cp}(24) + \text{ep}(72) + \text{co}(3) + \text{eo}(7)$
6	$1/3$	35	$V_{1/3}$	$\text{cp}(32) + \text{co}(3)$

Total: $20 + 2 + 39 + 26 + 106 + 35 = 228$. In the registered canonical block census, $k = 5$ ($\lambda = 4/9$) is absent from every block spectrum.

A.2 Block k -Sets

The union formula $\mathcal{K}(A) = \bigcup_B \mathcal{K}_B$ (Theorem 3.4):

Block	Dim	k -set	Algebraic origin
cp	64	$\{0, 4, 6\}$	Q_3 Hamming scheme (Krawtchouk)
ep	144	$\{0, 2, 3, 4\}$	Face-incidence adjacency algebra
co	8	$\{3, 4, 6\}$	Symmetry-guided computation
eo	12	$\{1, 2, 4\}$	Numerical-representation observation

Union: $\{0, 4, 6\} \cup \{0, 2, 3, 4\} \cup \{3, 4, 6\} \cup \{1, 2, 4\} = \{0, 1, 2, 3, 4, 6\}$. This is the six-value outcome of the registered block census. The absent values are not promoted here to a universal exclusion theorem.

Appendix B Related Work and Novelty Boundary

The closest mathematical backgrounds are association schemes and Bose–Mesner algebras for the permutation blocks, finite-group representation theory for the cubie representation, and harmonic analysis on finite groups for averaging operators. The core question is not a general commutative-operator theorem, but how exact block reductions and conditional arithmetic criteria relate to the displayed registered spectral values.

The association-scheme background is supplied by the standard texts of Bannai–Ito and Godsil [1, 6]. The CP block is the binary Hamming scheme $H(3, 2)$, whose Krawtchouk eigenvalues account for the CP block’s integer spectral contribution. The EP face-incidence algebra sits near the coherent-configuration side of the same algebraic-combinatorial framework.

The representation-theoretic background is finite semisimple representation theory [10, 4]. The averaging operator is also the representation-theoretic form of a random walk operator on a finite group, in the sense of Diaconis [5]. The Rubik group and cubie-state conventions follow the standard cube-group literature [7].

Rubik computational context. The declared 18-generator face-turn setting also occurs in the computational cube literature, including Kociemba’s two-phase algorithm and the diameter computation of Rokicki–Kociemba–Davidson–Dethridge [8, 9]. These references provide problem context rather than support for the spectral results.

Related work studies the finer QT/HT joint-spectral resolution and direct transport [3]. Extended tables and earlier numerical controls are available as supplementary material and are not used in the proofs.

Appendix C Computational Artifacts

The following repository artifacts provide the reproducible support layer for the registered finite computations in this paper. The default directory is `experiments/paper1/`; paths in the table are relative to that directory.

Artifact	Role	Short path
B1	registered six-layer spectrum and block census	<code>validation/spectral_ladder.py</code>
B2	registered absence of the $k = 5$ layer	<code>validation/k_absence.py</code>
B3	blockwise spectral support	<code>validation/block_composition.py</code>
B4	numerical projector identities and residuals	<code>validation/projector_algebra.py</code>
B5	qualified CO/EO block audit	<code>validation/co_eo_analytic_spectrum.py</code>
B6	finite broken-face arithmetic control	<code>validation/symmetry_breaking.py</code>
B7	frozen claim-aware figure display data	<code>results/figure_data.json</code>

From the repository root, an executable artifact is run as `python experiments/paper1/<short path>`. These artifacts support the stated Computational Certificates and Computational Observations; they are not used in the exact proofs.

All listed artifacts are available in the [RIME repository](#).

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